

# Malignant Infarction

Decompressive hemicraniectomy selection criteria, evidence, and supportive ICU care.

Monitor for deterioration and involve neurosurgery early. The evidence for decompression within 48 hours is a treatment window in selected patients, not a claim that edema always peaks at 24–48 hours. Dangerous swelling can occur early or later, often over days 3–5.

## 1. Surgical Assessment and Prognostic Markers

### Clinical Deficit Severity:

- Assess clinical deterioration, consciousness, infarct extent, age, premorbid function and goals with neurosurgery. A fixed hemisphere-specific NIHSS cutoff does not define universal eligibility.
- **Timing:** the strongest trial evidence concerns selected patients treated within 48 hours; evaluate early rather than waiting for a numeric gate.

### Imaging and predictive cohort markers (not standalone surgical eligibility):

- Infarction of  $\geq 50\%$  of the MCA territory (CT/MRI)
- DWI core volume  $> 82$  mL within 6 hours predicts a malignant course (prognostic cohort threshold: specificity 0.98, sensitivity 0.52)
- DWI core volume  $> 145$  mL within 14 hours
- Midline shift or mass effect on repeat imaging
- **Operative planning:** decompression technique and extent require a neurosurgical plan; no standalone bone-flap recipe is prescribed here.

## 2. Surgical Outcomes & Evidence (By Age Group)

### Age 18–60 years: pooled DECIMAL/DESTINY/HAMLET, one-year outcomes

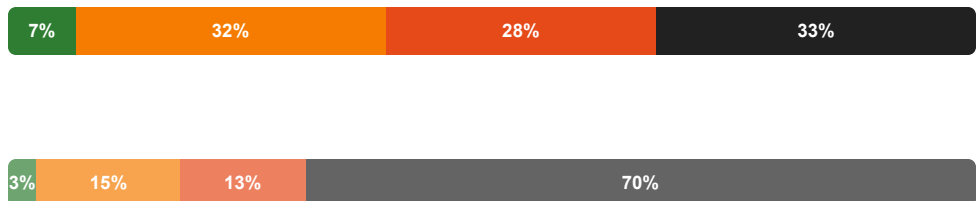
Survival was 78% with surgery versus 29% with medical care; mRS  $\leq 3$  was 43% versus 21%, and mRS  $\leq 4$  was 75% versus 24%. These are group outcomes from selected trial participants, not individual forecasts.

The separate mRS 4 and 5 plot segments are unavailable because their apparent derivation from rounded aggregates was not independently verified against the original figure.

### Age $\geq 61$ Years (DESTINY II; 6-month outcomes)

Surgery (33% Mort) vs Med (70% Mort)

Age  $\geq 61$  Medical Control



■ mRS 0–2: Functional independence; mRS 3: walks unassisted but needs some help

■ mRS 4: Moderately severe; unable to walk or attend bodily needs unassisted

■ mRS 5: Severe disability; bedridden / constant care ■ mRS 6: Death

- **Age 18–60:** pooled one-year NNT approximately 2 for survival and 4 for mRS  $\leq 3$ . **Age  $\geq 61$ :** DESTINY II improved survival without mRS 5–6 at six months; the mRS  $\leq 3$  comparison was not statistically significant. Displayed percentages are rounded; these are different study populations and follow-up horizons.

### 3. Supportive ICU Care & Medical Management

|                    |                                                                                                                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Positioning</b> | Elevate HOB 30 degrees; maintain straight head/neck alignment to maximize venous outflow.                                                                                                                                             |
| <b>Fluids</b>      | Maintain euvolemia with isotonic fluids. <b>Avoid hypotonic fluids</b> (e.g. D5W, 0.45% NS) that can worsen edema; balanced crystalloids such as LR should follow local neuro-ICU protocol.                                           |
| <b>Osmotherapy</b> | Consider <b>targeted PRN</b> hyperosmolar agents (HTS 3% or Mannitol) for acute decline or severe mass effect. <i>Prophylactic osmotherapy is not recommended.</i>                                                                    |
| <b>Metabolic</b>   | Assess and treat fever and ventilation abnormalities in clinical context. Avoid inappropriate hypocapnia; brief rescue hyperventilation for herniation is distinct from routine ventilation. Use a complete monitored treatment plan. |
| <b>Steroids</b>    | <b>Class III (Harmful):</b> Corticosteroids are NOT recommended for reducing cerebral edema in acute ischemic stroke.                                                                                                                 |

**DECIMAL:** Vahedi K et al. *Stroke*. 2007;38:2506-2517. **PMID: 17690311** (<https://pubmed.ncbi.nlm.nih.gov/17690311/>) | **DESTINY:** Jüttler E et al. *Stroke*. 2007;38:2518-2525. **PMID: 17690310** (<https://pubmed.ncbi.nlm.nih.gov/17690310/>)

**HAMLET:** Hofmeijer J et al. *Lancet Neurol*. 2009;8:326-333. **PMID: 19269254** (<https://pubmed.ncbi.nlm.nih.gov/19269254/>) | **DESTINY II:** Jüttler E et al. *N Engl J Med*. 2014;370:1091-1100. **PMID: 24645942** (<https://pubmed.ncbi.nlm.nih.gov/24645942/>)

**AHA Guidelines:** Wijdicks EF et al. *Stroke*. 2014;45:1222-1238. **PMID: 24481970** (<https://pubmed.ncbi.nlm.nih.gov/24481970/>)